



Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
Institut Teknologi Sepuluh Nopember

Laporan Akhir

Praktikum Jaringan Komputer

Vlan-Trunk-OSPF-MultiVendor

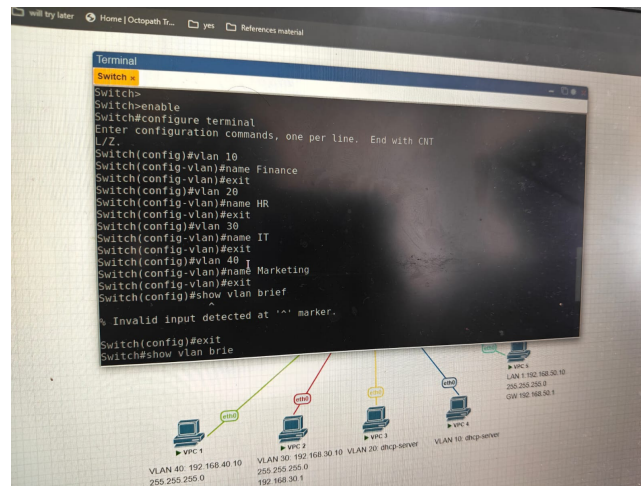
Muhammad Rasyha Syauqi Islam - 5024241066

2025

1 Langkah-Langkah Percobaan

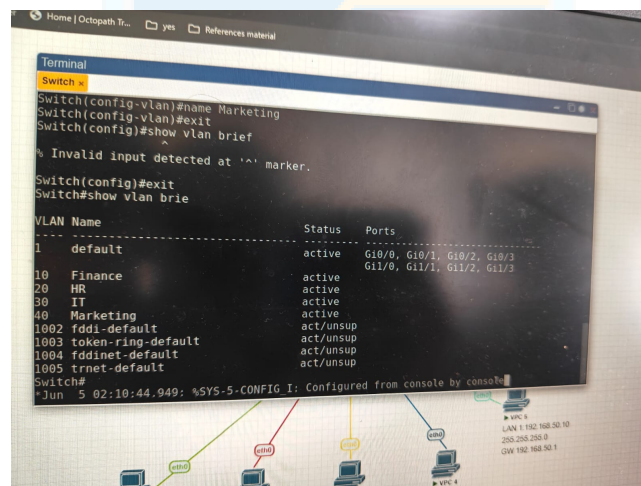
1.1 Praktikum 1 Konfigurasi Cisco Switch

- Masuk ke terminal Cisco Switch lalu
 - Switch> enable
 - Switch# configure terminal
 - Switch(config)#



Gambar 1: Membuat VLAN

- Memastikan vlan sudah dibuat dan statusnya aktif dengan cara **show vlan brief**



Gambar 2: Show Vlan Ip Brief

- Mengkonfigurasi Access port untuk setiap vlan

```
interface gi0/2
description PC-FINANCE-VLAN10
switchport mode access
switchport access vlan 10
no shutdown
```

```
exit
```

```
interface gi0/1  
description PC-HR-VLAN20  
switchport mode access  
switchport access vlan 20  
no shutdown  
exit
```

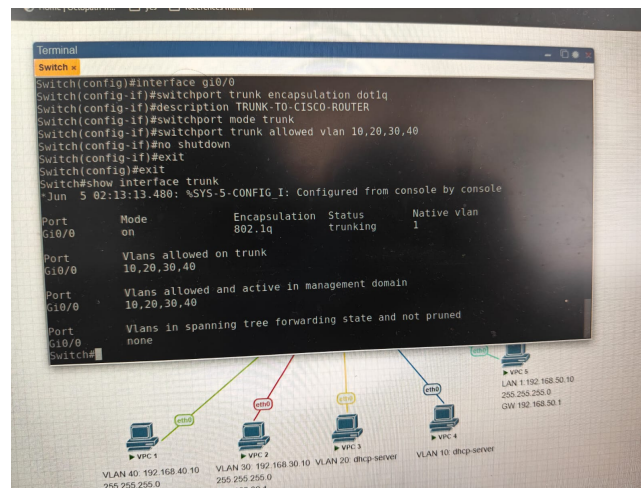
```
interface gi1/1  
description PC-IT-VLAN30  
switchport mode access  
switchport access vlan 30  
no shutdown  
exit
```

```
interface gi0/3  
description PC-MARKETING-VLAN40  
switchport mode access  
switchport access vlan 40  
no shutdown  
exit
```

- Mengkonfigurasi Trunk ke Cisco Router

```
interface gi0/0  
switchport trunk encapsulation dot1q  
description TRUNK-TO-CISCO-ROUTER  
switchport mode trunk  
switchport trunk allowed vlan 10,20,30,40  
no shutdown  
exit
```

Lalu verifikasi dengan
show interfaces trunk



Gambar 3: Show Interfaces Trunk

1.2 Praktikum 2 Konfigurasi Cisco Router

- Masuk ke Cisco Router
enable
configure terminal
- Konfigurasi Interface Trunk ke Switch, lalu pilih interface yang terhubung ke Cisco Switch. Pada topologi, interface yang terhubung adalah Gig0/2

```
interface gig0/2
description TRUNK-TO-CISCO-SWITCH
no ip address
no shutdown
exit
```

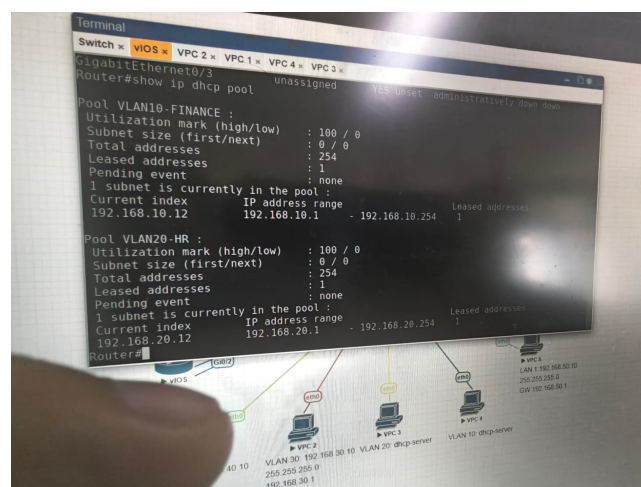
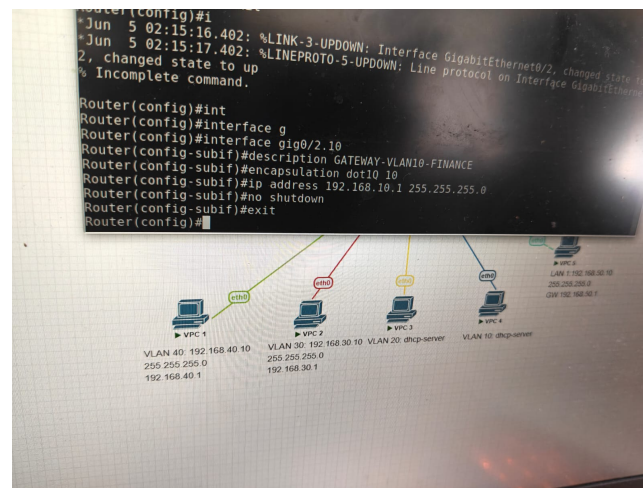
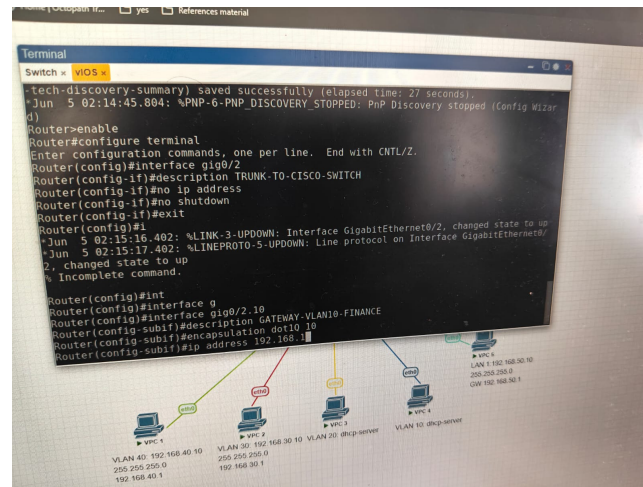
- Konfigurasi Subinterface VLAN 10
interface gig0/2.10
description GATEWAY-VLAN10-FINANCE
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
- Konfigurasi Subinterface VLAN 20
interface gig0/2.20
description GATEWAY-VLAN20-HR
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
no shutdown

exit

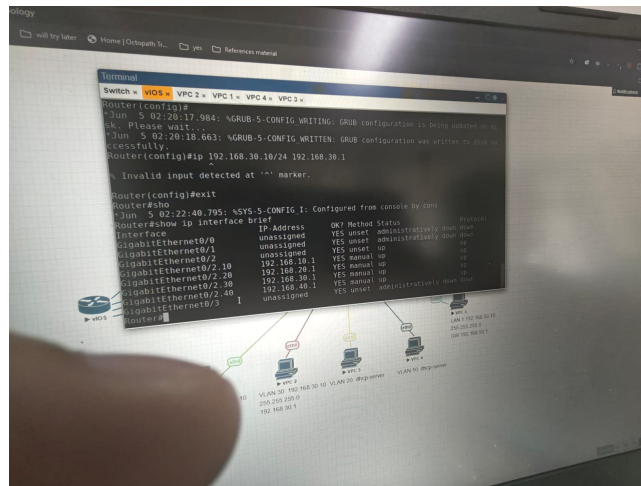
- Konfigurasi Subinterface VLAN 30
interface gig0/2.30
description GATEWAY-VLAN30-IT
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
no shutdown
exit
- Konfigurasi Subinterface VLAN 40
interface gig0/2.40
description GATEWAY-VLAN40-MARKETING
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
no shutdown
exit
- Konfigurasi DHCP-Server untuk Vlan 10
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp pool VLAN10-FINANCE
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
exit
- Konfigurasi DHCP-Server untuk Vlan 20
ip dhcp excluded-address 192.168.20.1 192.168.20.10
ip dhcp pool VLAN20-HR
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
exit
- Konfigurasi IP Static pada VPCS untuk vlan 30 dan 40
PC VLAN 30 : ip 192.168.30.10/24 192.168.30.1
PC Vlan 40 : ip 192.168.40.10/24 192.168.40.1
PC Vlan 10 dan 20 menggunakan dhcp: ip dhcp
- Verifikasi Cisco Router
Cek interface : show ip interface brief

Cek dhcp pool : show ip dhcp pool

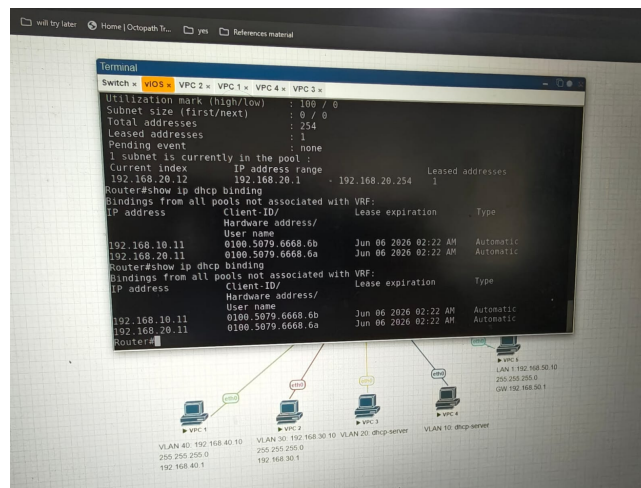
Cek client DHCP yang mendapatkan IP : show ip dhcp binding



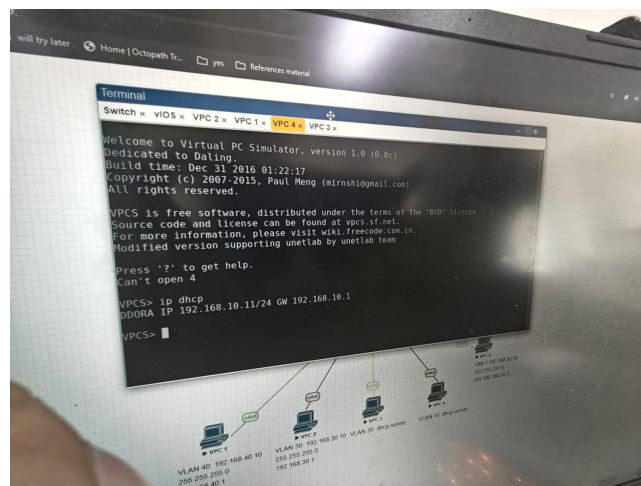
Gambar 4: Show Ip Dhcp Pool



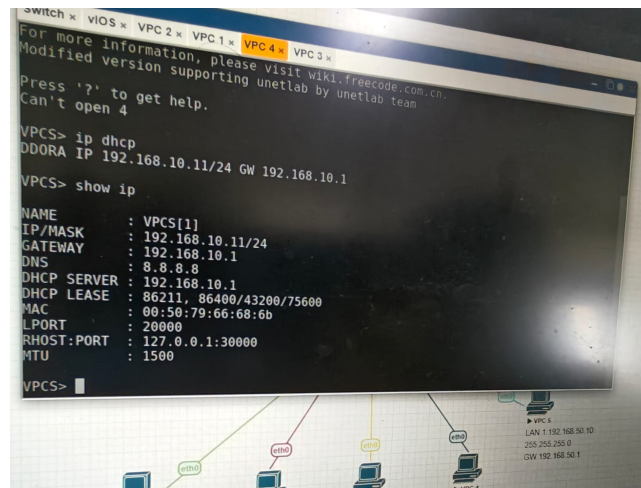
Gambar 5: Show IP Int Brief



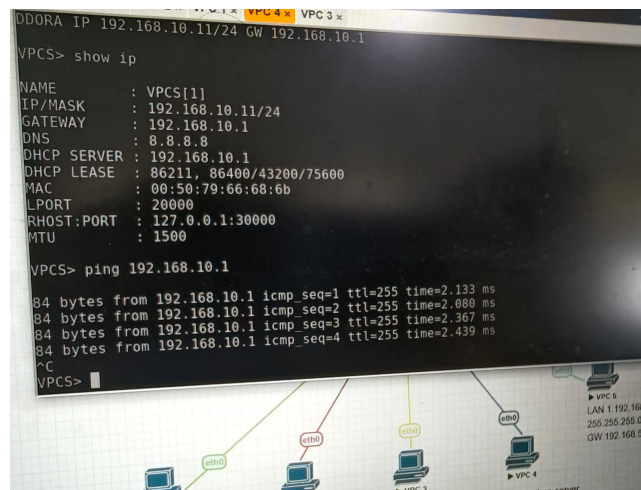
- Pengujian dari VPCS
 - Masuk ke pc vlan 10 untuk request ip secara dhcp : ip dhcp



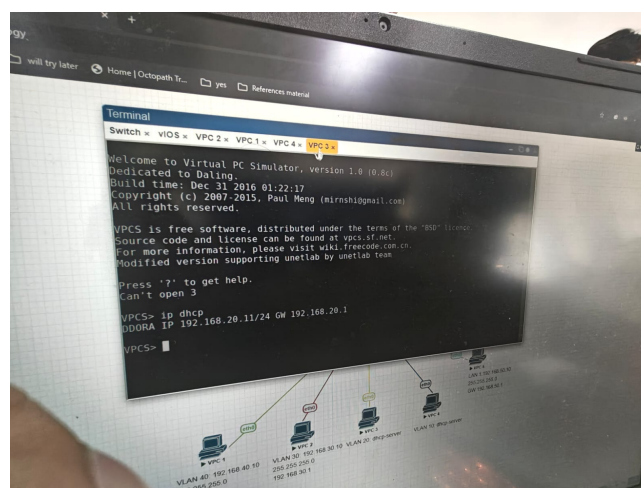
Pastikan pc vlan 10 sudah dapat ip secara dhcp dengan command : show ip



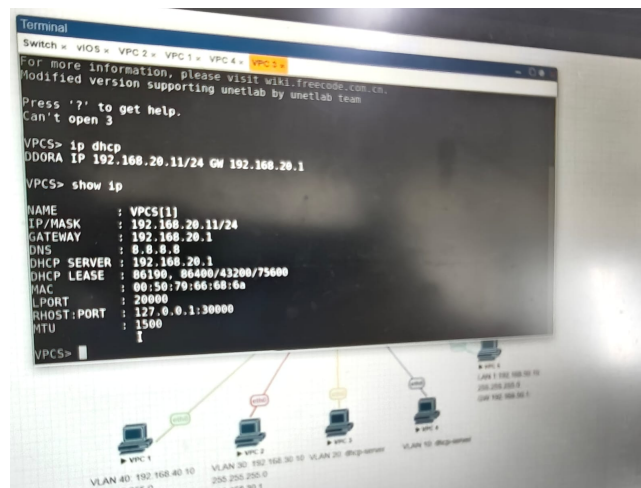
Dari pc vlan 10 ping gateway 192.168.10.1 : ping 192.168.10.1



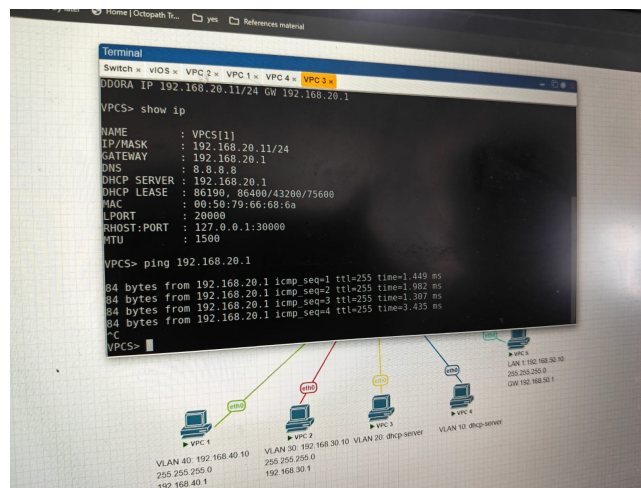
Masuk ke pc vlan 20 untuk request ip secara dhcp : ip dhcp



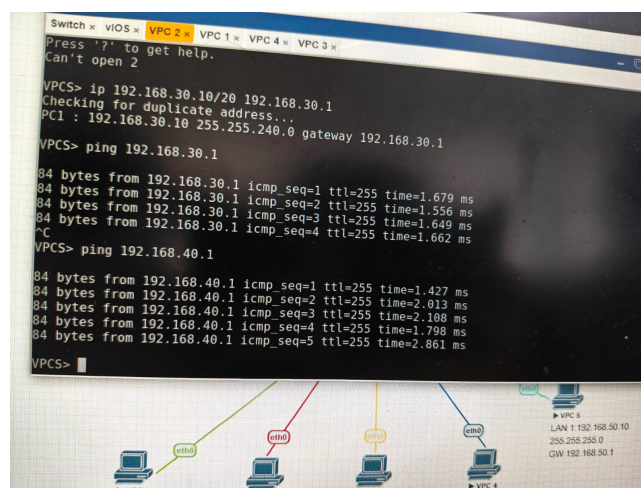
Pastikan pc vlan 20 sudah dapat ip secara dhcp dengan command : show ip



Dari pc vlan 20 ping gateway 192.168.20.1 : ping 192.168.20.1



Dari pc vlan 30 ping gateway 192.168.30.1 : ping 192.168.30.1



Dari pc vlan 40 ping gateway 192.168.40.1 : ping 192.168.40.1

Uji ping antar-Vlan dari vlan 10 :

ping 192.168.20.11

ping 192.168.30.10

ping 192.168.40.10

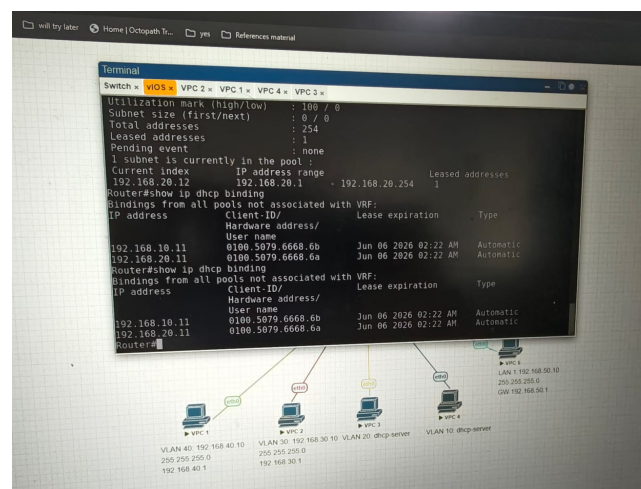
```
VPCS> ping 192.168.20.11
84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=11.996 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=2.872 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=3.159 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=2.528 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=1.991 ms

VPCS> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=17.326 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=3.582 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=2.927 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=3.844 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=3.173 ms

VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=16.969 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=2.193 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=2.431 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=2.373 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=3.438 ms

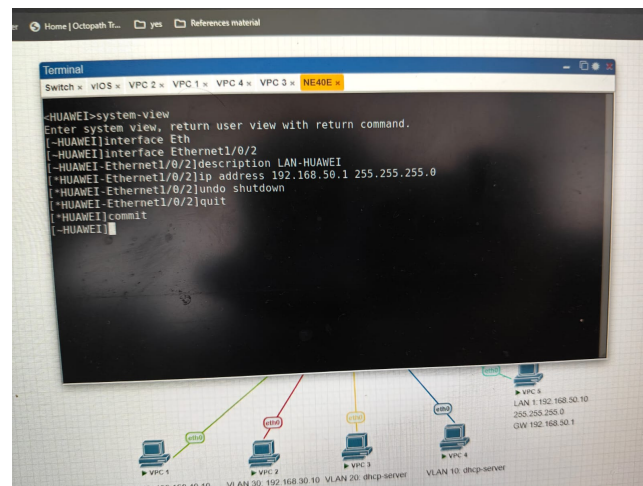
VPCS>
```

Masuk ke cisco router lagi untuk melihat ip dhcp binding : show ip dhcp binding

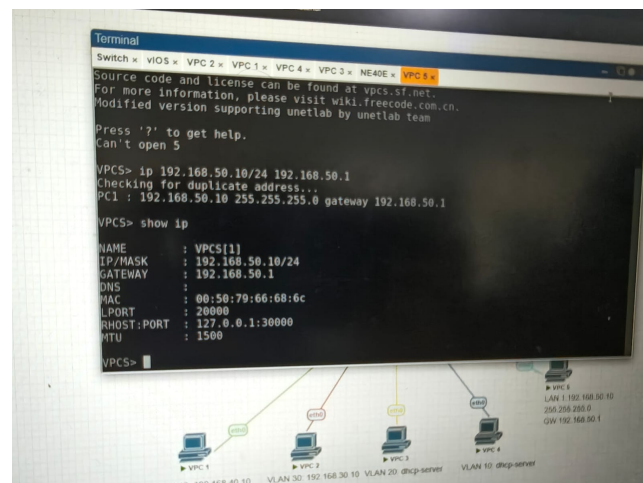


1.3 Praktikum 3 Konfigurasi Huawei Router dan IP Address Antarrouter

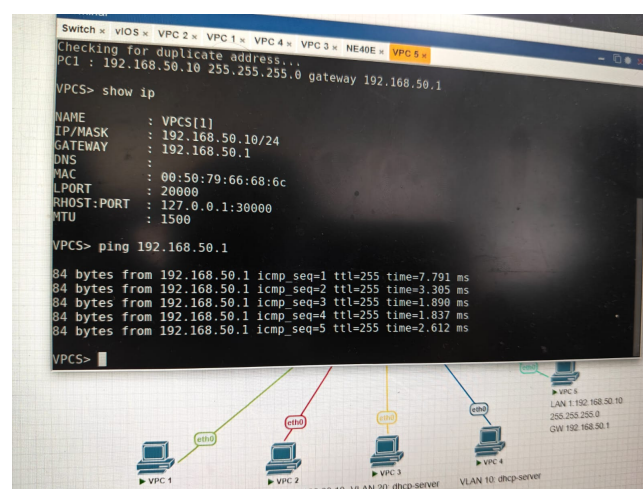
- Konfigurasi Lan Huawei
konfigurasi ip address untuk lan huawei:
system-view
interface Ethernet1/0/2
description LAN-HUAWEI
ip address 192.168.50.1 255.255.255.0
undo shutdown
quit
commit



- Konfigurasi IP VPC Lan Huawei
 Pada pc lan huawei: ip 192.168.50.10/24 192.168.50.1
 Verifikasi : show ip



- Testing ping gateway huawei ping: ping 192.168.50.1



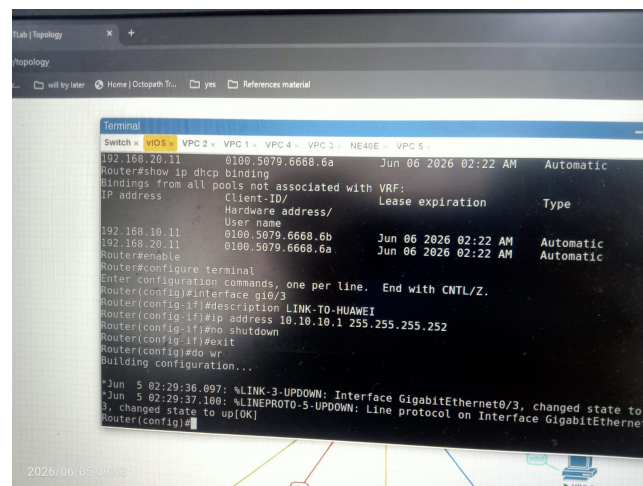
- Konfigurasi IP link cisco <-> Huawei link 1

Tabel 1: Konfigurasi IP Address Antar Router

Perangkat	Interface	IP Address
Cisco Router	Gi0/3	10.10.10.1/30
Huawei Router	Ethernet1/0/0	10.10.10.2/30

Cisco Router

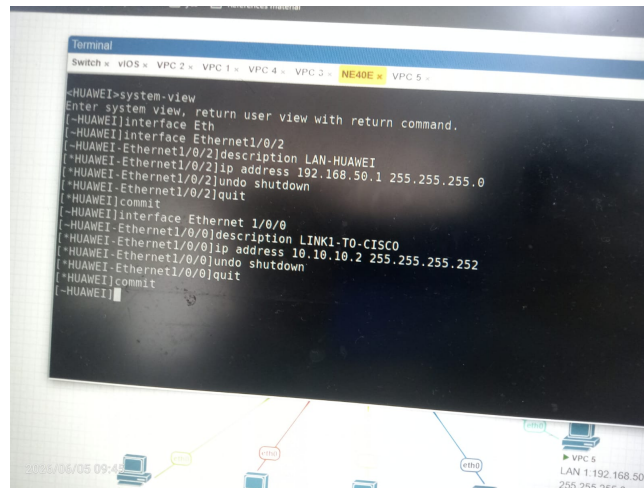
enable
configure terminal
interface gi0/3
description LINK1-TO-HUAWEI
ip address 10.10.10.1 255.255.255.252
no shutdown
exit



```
Switch > VIOS > VPC 2 > VPC 1 > VPC 4 > VPC 3 > NE40E > VPC 5
192.168.20.11 0100.5079.6668.6a Jun 06 2026 02:22 AM Automatic
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address Client-ID/ Lease expiration Type
Hardware address/
User name
192.168.20.11 0100.5079.6668.6b Jun 06 2026 02:22 AM Automatic
192.168.20.11 0100.5079.6668.6a Jun 06 2026 02:22 AM Automatic
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gi0/3
Router(config-if)#description LINK-TO-HUAWEI
Router(config-if)#ip address 10.10.10.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#do wr
Building configuration...
*Jun 5 02:29:36.007: %LINK-3-UPDOWN: Interface GigabitEthernet0/3, changed state to u
*Jun 5 02:29:37.100: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0
Router(config)#
```

Huawei Router

system-view
interface Ethernet1/0/0
description LINK1-TO-CISCO
ip address 10.10.10.2 255.255.255.252
undo shutdown
quit
commit



- Konfigurasi IP Link Cisco <-> Huawei Link 2

Tabel 2: Konfigurasi IP Address Antar Router

Perangkat	Interface	IP Address
Cisco Router	Gi0/0	10.10.11.1/30
Huawei Router	Ethernet1/0/4	10.10.11.2/30

Cisco Router

```
interface gi0/0
description LINK2-TO-HUAWEI
ip address 10.10.11.1 255.255.255.252
no shutdown
exit
```

Huawei Router

```
system-view
interface Ethernet1/0/4
description LINK2-TO-CISCO
ip address 10.10.11.2 255.255.255.252
undo shutdown
quit
commit
```

- Konfigurasi IP link Huawei <-> Mikrotik

Tabel 3: Konfigurasi IP Address Antar Router

Perangkat	Interface	IP Address
Huawei Router	Ethernet1/0/3	10.10.20.1/30
MikroTik	ether1	10.10.20.2/30

Huawei Router

```
system-view
interface Ethernet1/0/3
description LINK-TO-MIKROTIK
```


ip address 10.10.20.1 255.255.255.252

undo shutdown

quit

Mikrotik

/ip address add address=10.10.20.2/30 interface=ether1 comment=TO-HUAWEI

commit

- Konfigurasi IP link Cisco <-> Mikrotik

Tabel 4: Konfigurasi IP Address Antar Router

Perangkat	Interface	IP Address
Cisco Router	Gi0/1	10.10.30.1/30
MikroTik	ether2	10.10.30.2/30

Cisco Router

interface gi0/1

description LINK-TO-MIKROTIK

ip address 10.10.30.1 255.255.255.252

no shutdown

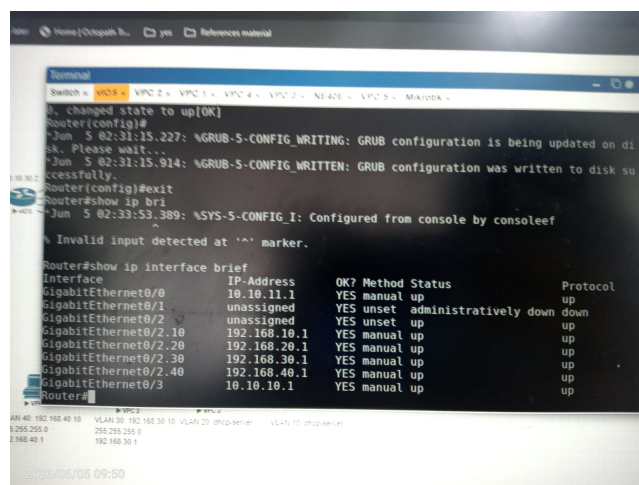
exit

Mikrotik

/ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO

- Verifikasi Interface

Cisco Router : show ip interface brief



Huawei Router : display ip interface brief


```

Terminal
Switch > vIOS > VPC2 > VPC1 > VPC4 > VPC3 > NE40E > VPC5 > Mikrotik >
rdown: FIB overload down
down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface          IP Address/Mask    Physical    Protocol    VPN
-----
Ethernet1/0/0      10.10.10.2/30       up          up          ..
Ethernet1/0/0.4094 128.1.138.137/16    up          up          13vpn
Ethernet1/0/1       unassigned          up          down         ..
Ethernet1/0/1.4094 128.1.138.137/16    up          up          13vpn
Ethernet1/0/2       192.168.50.1/24     up          up          ..
Ethernet1/0/2.4094 128.1.138.137/16    up          up          13vpn
Ethernet1/0/3       10.10.20.1/30       up          up          ..
Ethernet1/0/3.4094 128.1.138.137/16    up          up          13vpn
Ethernet1/0/4       10.10.11.2/30       up          up          ..
Ethernet1/0/4.4094 128.1.138.137/16    up          up          13vpn
... More ...

```

Mikrotik : /ip address print

```

Terminal
Switch > vIOS > VPC2 > VPC1 > VPC4 > VPC3 > NE40E > VPC5 > Mikrotik >
101 101 111 KKKKK RRR RRR 000 000 TTT III KKKKK
101 101 111 KKK KKK RRRRR 000 000 TTT III KKK KKK

do you want to see the software license? [Y/n]: n
[?] Gives the list of available commands
command [?] Gives help on the command and list of arguments
[Tab] Completes the command/word. If the input is ambiguous,
a second [Tab] gives possible options
Move up to base level
Move up one level
Command Use command at the base level
Jun/05/2026 02:09:55 system,error,critical router was rebooted without proper shu
tdown
Change your password
[admin@mikrotik] > /ip address add address=10.10.20.2/30 interface=ether1 comment=T0-HU
[admin@mikrotik] > 10.10.20.0 ether1

```

- Verifikasi ping Antar Router
- Dari Cisco Router :
- ping 10.10.10.2
- ping 10.10.11.2
- ping 10.10.30.2

```

Terminal
Switch > vIOS > VPC2 > VPC1 > VPC4 > VPC3 > NE40E > VPC5 > Mikrotik >
Jun 5 02:42:00.345: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/
3, changed state to up
Router#show ip int brief
Interface          IP-Address         OK? Method Status    Protocol
GigabitEthernet0/0 10.10.11.1         YES manual up        up
GigabitEthernet0/1 10.10.30.1         YES manual up        up
GigabitEthernet0/2  unassigned        YES unset up        up
GigabitEthernet0/2.10 192.168.10.1     YES manual up        up
GigabitEthernet0/2.20 192.168.20.1     YES manual up        up
GigabitEthernet0/2.30 192.168.30.1     YES manual up        up
GigabitEthernet0/2.40 192.168.40.1     YES manual up        up
GigabitEthernet0/3  10.10.10.1        YES manual up        up
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/2/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#show ip int brief

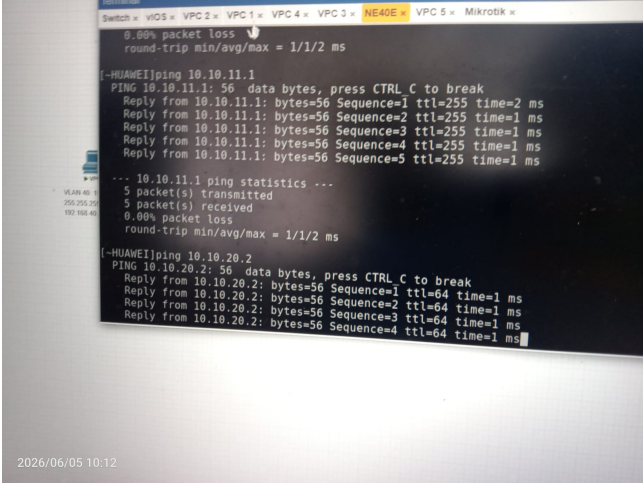
```

Dari Router Huawei :

ping 10.10.10.1

ping 10.10.11.1

ping 10.10.20.2



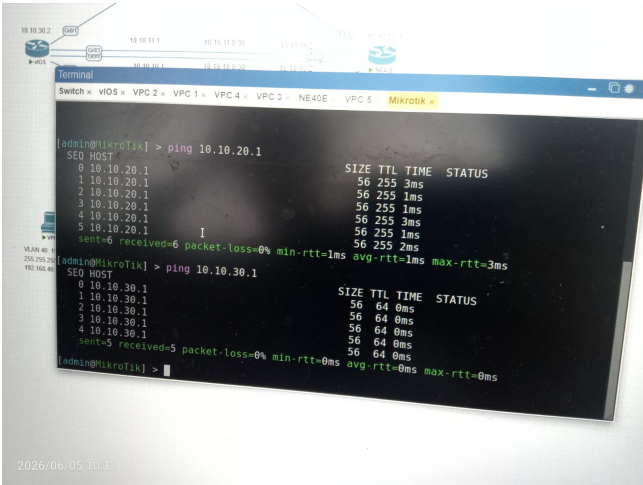
```
Switch > VIOS > VPC2 > VPC1 > VPC4 > VPC3 > NE40E > VPC5 > Mikrotik >  
0.00% packet loss  
round-trip min/avg/max = 1/1/2 ms  
[~HUAWEI]ping 10.10.11.1  
PING 10.10.11.1: 56 data bytes, press CTRL C to break  
Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=2 ms  
Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=1 ms  
Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=1 ms  
Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=1 ms  
Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=1 ms  
--- 10.10.11.1 ping statistics ---  
5 packet(s) transmitted  
5 packet(s) received  
0.00% packet loss  
round-trip min/avg/max = 1/1/2 ms  
[~HUAWEI]ping 10.10.20.2  
PING 10.10.20.2: 56 data bytes, press CTRL C to break  
Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms  
Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms  
Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms  
Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
```

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Dari Router Mikrotik :

ping 10.10.20.1

ping 10.10.30.1



```
Switch > VIOS > VPC2 > VPC1 > VPC4 > VPC3 > NE40E > VPC5 > Mikrotik >  
[admin@MikroTik] > ping 10.10.20.1  
SEQ HOST      SIZE TTL TIME STATUS  
0 10.10.20.1  56 255 3ms  
1 10.10.20.1  56 255 1ms  
2 10.10.20.1  56 255 1ms  
3 10.10.20.1  56 255 1ms  
4 10.10.20.1  56 255 3ms  
5 10.10.20.1  56 255 1ms  
6 10.10.20.1  56 255 2ms  
sent=6 received=6 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=3ms  
[admin@MikroTik] > ping 10.10.30.1  
SEQ HOST      SIZE TTL TIME STATUS  
0 10.10.30.1  56 64 0ms  
1 10.10.30.1  56 64 0ms  
2 10.10.30.1  56 64 0ms  
3 10.10.30.1  56 64 0ms  
4 10.10.30.1  56 64 0ms  
5 10.10.30.1  56 64 0ms  
sent=5 received=5 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=0ms  
[admin@MikroTik] >
```

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1.4 Praktikum 4 Konfigurasi OSPF Multivendor dan OSPF Cost

Konsep routing ospf pada topologi.

Pada topologi ini, cisco router dan huawei memiliki 2 jalur utama. Lalu, mikrotik digunakan sebagai jalur cadangan jika 2 link utama tidak berfungsi.

Tabel 5: Konfigurasi Cost dan Fungsi Link

Link	Cost	Fungsi
Cisco ↔ Huawei Link 1	10	Jalur utama
Cisco ↔ Huawei Link 2	10	Jalur utama
Cisco ↔ MikroTik	100	Jalur backup
Huawei ↔ MikroTik	100	Jalur backup

Network yang diiklankan ke ospf

Cisco Router

Tabel 6: Daftar Network dan Keterangan Router Cisco

Network	Keterangan
10.10.10.0/30	Link Cisco-Huawei 1
10.10.11.0/30	Link Cisco-Huawei 2
10.10.30.0/30	Link Cisco-MikroTik
192.168.10.0/24	VLAN 10 Finance
192.168.20.0/24	VLAN 20 HR
192.168.30.0/24	VLAN 30 IT
192.168.40.0/24	VLAN 40 Marketing

Huawei Router

Tabel 7: Daftar Network dan Keterangan Router Huawei

Network	Keterangan
10.10.10.0/30	Link Huawei-Cisco 1
10.10.11.0/30	Link Huawei-Cisco 2
10.10.20.0/30	Link Huawei-MikroTik
192.168.50.0/24	LAN Huawei

Mikrotik

Tabel 8: Daftar Network dan Keterangan Router MikroTik

Network	Keterangan
10.10.20.0/30	Link MikroTik-Huawei
10.10.30.0/30	Link MikroTik-Cisco

- Konfigurasi ospf pada Cisco Router

Masuk ke Cisco Router:

```
enable  
configure terminal
```

Konfigurasi OSPF:

```
router ospf 1  
router-id 1.1.1.1  
network 10.10.10.0 0.0.0.3 area 0
```

```
network 10.10.11.0 0.0.0.3 area 0
network 10.10.30.0 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0
exit
```

Atur OSPF cost pada interface Cisco :

```
interface gi0/3
description LINK1-TO-HUAWEI
ip ospf cost 10
exit
```

```
interface gi0/0
description LINK2-TO-HUAWEI
ip ospf cost 10
exit
```

```
interface gi0/1
description LINK-TO-MIKROTIK
ip ospf cost 100
exit
```

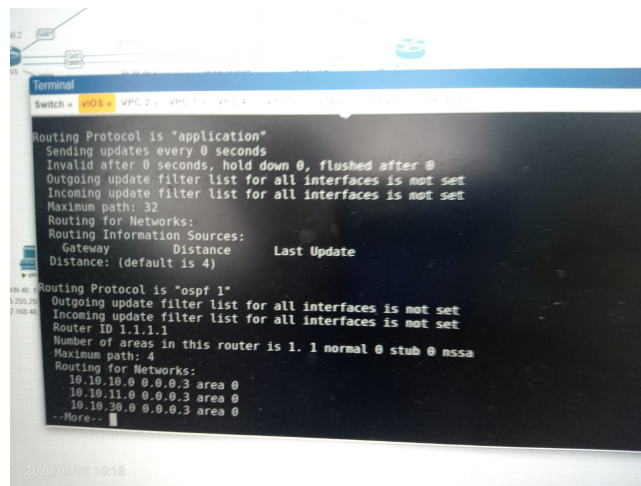
Verifikasi OSPF Cisco :

```
show ip ospf neighbor
show ip route ospf
```

Hasil kemungkinan kosong karena router tetangga masih belum dikonfigurasi : show ip protocols

Jika ip vlan tidak muncul, simpan konfigurasi cisco router dengan perintah wr, lalu ketik perintah reload. Tunggu proses restart sampai selesai, lalu cek lagi.

Pastikan network yang akan diiklankan sudah ada



- Konfigurasi OSPF pada Huawei Router
Masuk ke huawei router : system-view

Konfigurasi OSPF :

```
ospf 1 router-id 2.2.2.2
area 0.0.0.0
network 10.10.10.0 0.0.0.3
network 10.10.11.0 0.0.0.3
network 10.10.20.0 0.0.0.3
network 192.168.50.0 0.0.0.255
quit
quit
```

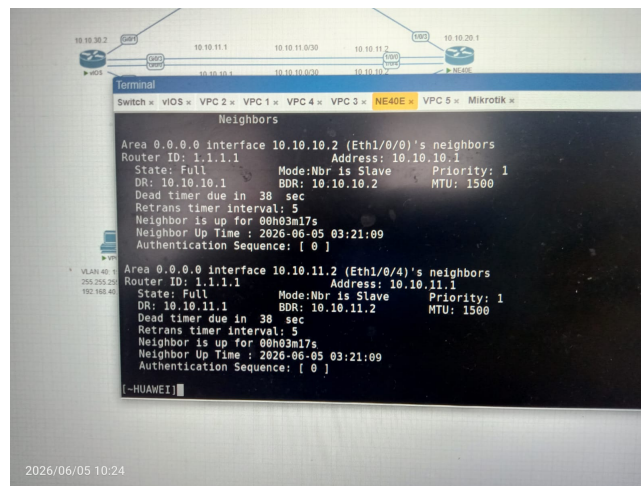
Atur OSPF cost pada interface huawei :

```
interface Ethernet1/0/0
description LINK1-TO-CISCO
ospf cost 10
quit
```

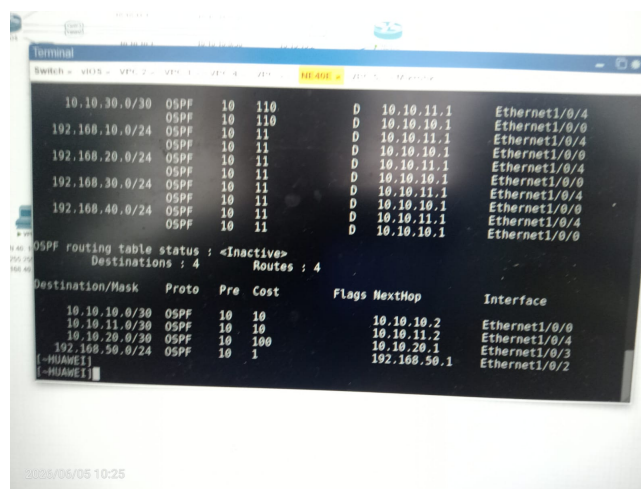
```
interface Ethernet1/0/4
description LINK2-TO-CISCO
ospf cost 10
quit
```

```
interface Ethernet1/0/3
description LINK-TO-MIKROTIK
ospf cost 100
quit
commit
```

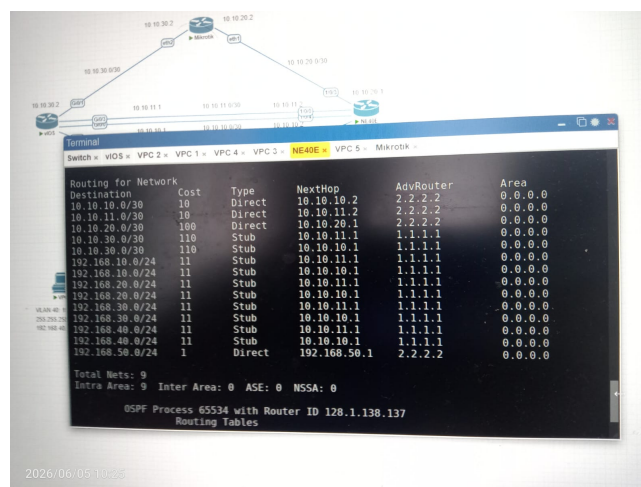
Verifikasi OSPF Huawei : display ospf peer



Pastikan huawei peer dengan cisco melalui 2 link utama : display ip routing-table protocol ospf
Pastikan hauwei sudah punya mendapatkan network lengkap dari cisco.



```
display ospf routing
```



- Konfigurasi OSPF pada MikroTik Pada topologi ini, Mikrotik digunakan sebagai jalur backup antara cisco dan huawei. Network Mikrotik :

Tabel 9: Konfigurasi Interface dan IP Address Router MikroTik

Link	Interface MikroTik	IP Address
MikroTik Huawei	ether1	10.10.20.2/30
MikroTik Cisco	ether2	10.10.30.2/30

Masuk ke terminal Mikrotik, lalu konfigurasi router-id OSPF : /routing ospf instance set default router-id=3.3.3.3

Tambahkan network OSPF:

/routing ospf network add network=10.10.20.0/30 area=backbone

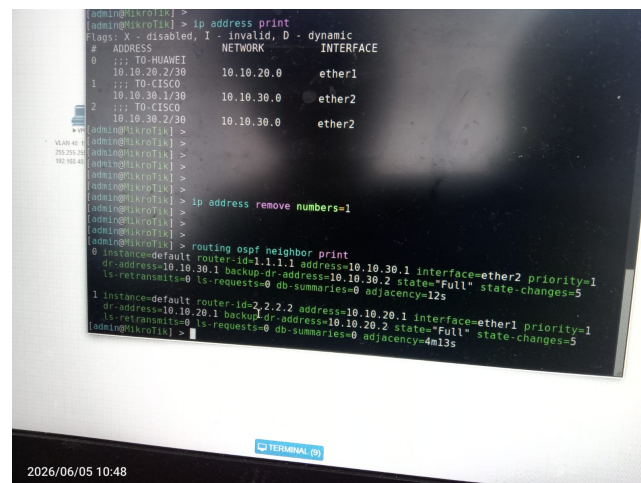
/routing ospf network add network=10.10.30.0/30 area=backbone

Atur cost interface mikrotik menjadi jalur bacakup :

/routing ospf interface add interface=ether1 cost=100

/routing ospf interface add interface=ether2 cost=100

Verifikasi Mikrotik : /routing ospf neighbor print



Pastikan router id cisco dan huawei sudah muncul dimikrotik

Cek routing table OSPF: /ip route print where ospf

Hasil yang diharapkan: MikroTik mendapatkan route OSPF menuju VLAN 10, 20, 30, 40 dan LAN Huawei 192.168.50.0/24.

```

[admin@MikroTik] > /routing ospf instance set default router-id=3.3.3.3
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print
0 instance-default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1,
dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=75
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
- blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC  GATEWAY      DISTANCE
0 Ado 10.10.10.0/30      10.10.20.1    110
1 Ado 10.10.11.0/30      10.10.20.1    110
2 Ado 192.168.10.0/24     10.10.20.1    110
3 Ado 192.168.20.0/24     10.10.20.1    110
4 Ado 192.168.30.0/24     10.10.20.1    110
5 Ado 192.168.40.0/24     10.10.20.1    110
6 Ado 192.168.50.0/24     10.10.20.1    110
[admin@MikroTik] >

```

- Verifikasi OSPF Neighbor
Cisco Router : show ip ospf neighbor
Hasil yang diharapkan:
Neighbor Huawei muncul dua kali.
Neighbor MikroTik muncul satu kali.

```

Router#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:39 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/BDR 00:00:35 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/BDR 00:00:38 10.10.10.2 GigabitEthernet0/3
Router#

```

- Huawei Router : display ospf peer
Hasil yang diharapkan:
Peer ospf muncul dua kali.
Peer mikrotik muncul satu kali.

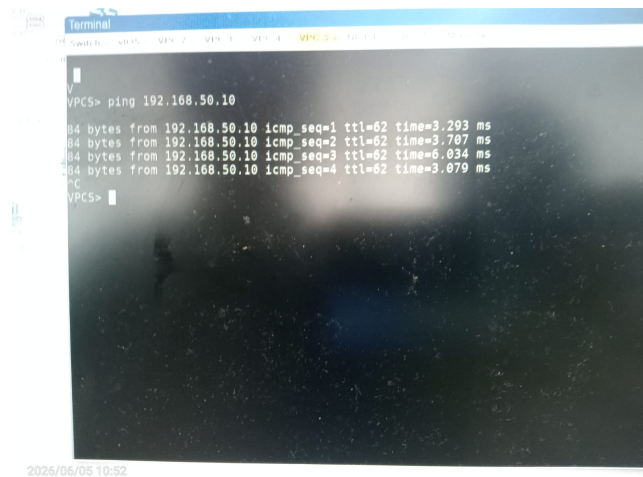
```

Neighbors
Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 38 sec
Retrans timer interval: 5
Neighbor is up for 00h03m17s
Neighbor Up Time : 2026-06-05 03:21:09
Authentication Sequence: [ 0 ]

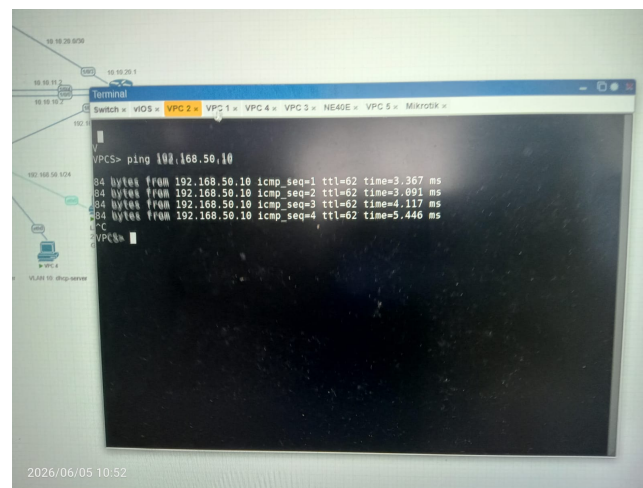
Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 38 sec
Retrans timer interval: 5
Neighbor is up for 00h03m17s
Neighbor Up Time : 2026-06-05 03:21:09
Authentication Sequence: [ 0 ]
[~HUAWEI~]

```

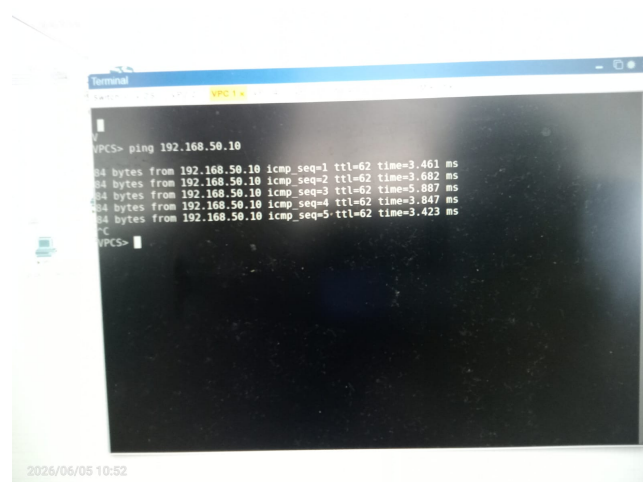
- Mikrotik : /routing ospf neighbor print
Hasil yang diharapkan:
Neighbor cisco muncul.
Neighbor Huawei muncul.



Dari pc vlan 30 : ping 192.168.50.10



Dari pc vlan 40 : ping 192.168.50.10



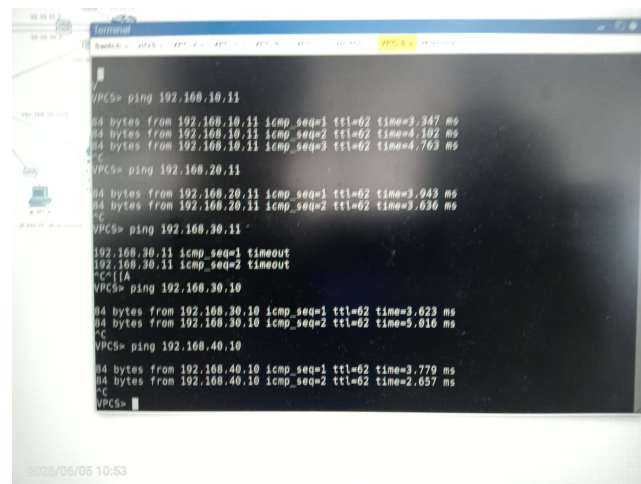
Lakukan pengujian dari pc lan huawei ke ip vlan cisco:

ping 192.168.10.11

ping 192.168.20.11

ping 192.168.30.10

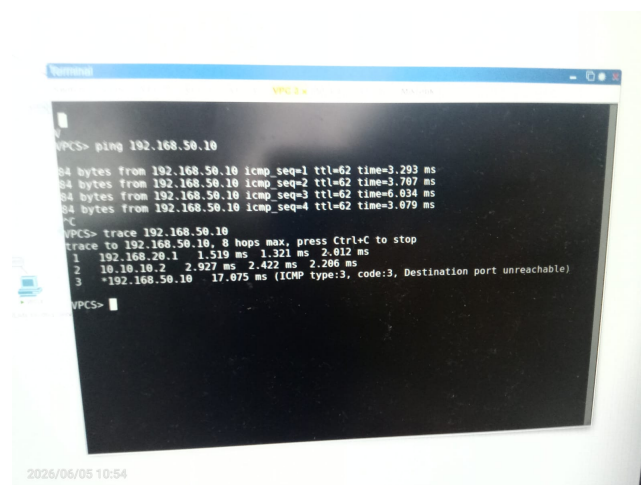
ping 192.168.40.10



```
PC> ping 192.168.10.11
64 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=3.347 ms
64 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=4.192 ms
64 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=4.763 ms
PC> ping 192.168.20.11
64 bytes from 192.168.20.11 icmp_seq=1 ttl=62 time=3.943 ms
64 bytes from 192.168.20.11 icmp_seq=2 ttl=62 time=3.636 ms
PC> ping 192.168.30.11
192.168.30.11 icmp_seq=1 timeout
192.168.30.11 icmp_seq=2 timeout
PC> ping 192.168.30.10
64 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=3.623 ms
64 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=5.016 ms
PC> ping 192.168.40.10
64 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=3.779 ms
64 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=2.657 ms
PC>
```

1.5 Praktikum 5 Pengujian Failover OSPF

- Kondisi normal: Dua link cisco-huawei aktif
Cek route dari cisco menuju lan huawei: show ip route 192.168.50.0
Lakukan trace dari pc vlan menuju lan huawei : trace 192.168.50.10
Hasil yang diharapkan : Traffic melewati jalur langsung Cisco → Huawei.



```
PC> ping 192.168.50.10
64 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.293 ms
64 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.707 ms
64 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.034 ms
64 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.079 ms
PC> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1 192.168.20.1 1.519 ms 1.321 ms 2.012 ms
 2 192.168.10.2 2.927 ms 2.422 ms 2.206 ms
 3 192.168.50.10 17.075 ms (ICMP type:3, code:3, Destination port unreachable)
PC>
```

- Kondisi 1 link cisco-huawei mati
Matikan salah satu link cisco-huawei.
configure terminal
interface gi0/3
shutdown
exit
Cek route dari cisco menuju lan huawei : show ip route 192.168.50.0
Lakukan trace dari pc vlan menuju lan huawei : trace 192.168.50.10
Hasil yang diharapkan: Route berpindah ke jalur backup melalui MikroTik.
Jalur: PC VLAN → Cisco Router → MikroTik → Huawei Router → LAN Huawei



Gambar 6: Caption

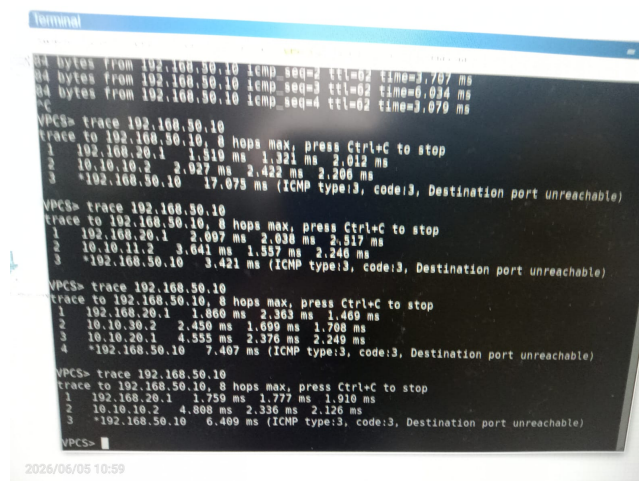
- Mengaktifkan kembali dua link cisco huawei
configure terminal
interface gi0/3
no shutdown
exit

interface gi0/0
no shutdown
exit

Cek route dari cisco menuju lan huawei : show ip route 192.168.50.0

Lakukan trace dari pc vlan menuju lan huawei : trace 192.168.50.10

Hasil yang diharapkan : Jalur routing kembali pindah ke link utama cisco-huawei.



2 Analisis Hasil Percobaan

Berdasarkan hasil praktikum yang telah dilakukan, konfigurasi jaringan pada perangkat Cisco, Huawei, dan MikroTik berhasil diterapkan dengan baik. Pembuatan VLAN pada switch Cisco dan konfigurasi inter VLAN routing pada router Cisco memungkinkan setiap VLAN saling berkomunikasi sesuai

dengan konsep Router on a Stick. Konfigurasi DHCP juga berjalan dengan baik, dibuktikan dengan perangkat pada VLAN 10 dan VLAN 20 yang berhasil memperoleh alamat IP secara otomatis. Selain itu, koneksi antar router berhasil dibangun menggunakan alamat IP point-to-point sehingga seluruh perangkat dapat saling terhubung melalui pengujian ping. Penerapan routing dinamis OSPF multi-vendor berhasil membentuk hubungan neighbor dan mendistribusikan informasi routing antar router. Pengaturan cost OSPF menjadikan dua link Cisco-Huawei sebagai jalur utama dan MikroTik sebagai jalur cadangan. Pada pengujian failover, ketika salah satu link utama dimatikan, jalur komunikasi secara otomatis berpindah ke jalur backup melalui MikroTik dan kembali menggunakan jalur utama setelah koneksi dipulihkan. Hasil tersebut menunjukkan bahwa seluruh konfigurasi telah sesuai dengan teori dan tujuan praktikum berhasil tercapai, meskipun ketelitian dalam memasukkan konfigurasi sangat diperlukan agar tidak terjadi kesalahan yang dapat mengganggu konektivitas jaringan.

3 Hasil Tugas Modul

Telah Di Upload Di ReadMe Github

4 Kesimpulan

Berdasarkan hasil praktikum yang telah dilakukan, dapat disimpulkan bahwa konfigurasi VLAN, inter VLAN routing, DHCP, serta koneksi antar router Cisco, Huawei, dan MikroTik berhasil diterapkan dengan baik. Penerapan routing dinamis OSPF multivendor memungkinkan seluruh jaringan saling bertukar informasi routing dan berkomunikasi dengan lancar. Pengaturan nilai cost OSPF juga berhasil membedakan jalur utama dan jalur cadangan, sehingga ketika salah satu link mengalami gangguan, proses failover dapat berlangsung secara otomatis tanpa memutus konektivitas jaringan.

5 Lampiran

5.1 Dokumentasi saat praktikum